

Problem 3A: Review of the Ideal Gas and Maxwell Equations

Starting with the enthalpy differential derived in lecture:

$$(1) \quad dH = C_p dT + V(1 - \beta T) dP$$

holding dH constant, dH = 0

$$0 = C_p dT + V(1 - \beta T) dP$$

Rearrange for dT:

$$(2) \quad dT = \left(\frac{V}{C_p} (\beta T - 1) dP \right)$$

- (3) Take the partial derivative with respect to P while holding H constant. Then divide both sides by V and multiply both sides by C_p:

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = (\beta T - 1)$$

- (4) Using $\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$ derived in lecture:

- (5) Substitute equation for β into expression from step 3:

$$(6) \quad \rho C_p \left(\frac{\partial T}{\partial P} \right)_H = \left(\frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P T - 1 \right)$$

- (7) Rearrange the right-hand side of equation 6 so that:

$$\left(\frac{\partial V}{V} \cdot \frac{T}{\partial T} \right)_P - 1 = \left(\frac{\frac{\partial V}{V}}{\left(\frac{\partial T}{T} \right)_P} \right)_P - 1$$

- (8) If we integrate the $\left(\frac{\partial V}{V} \right)_P$ term from above: $\int \left(\frac{\partial V}{V} \right)_P = \ln V$

- (9) If we integrate the $\left(\frac{\partial T}{T} \right)_P$ term from above: $\int \left(\frac{\partial T}{T} \right)_P = \ln T$

- (10) Then to undo the integration, differentiate expressions 8 and 9:

$$\left(\frac{\partial V}{V} \right)_P = \partial \ln V$$

$$\left(\frac{\partial T}{T} \right)_P = \partial \ln T$$

- (11) Substituting the relation from step 10 into step 7 we prove:

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = \left(\frac{\partial \ln V}{\partial \ln T} \right)_P - 1$$

Part 2: Using the formula you derived calculate the temperature change of an ideal gas that is undergoing an adiabatic expansion through a throttling valve.

- (12) Using the ideal gas law PV=RT

Rearrange for V:

$$V = \frac{RT}{P}$$

- (13) Now take the derivative of the above relation with respect to T at constant P:

$$\left(\frac{\partial V}{\partial T} \right)_P = \frac{R}{P}$$

- (14) Multiply both sides of expression 13 by (T/V):

$$\frac{T}{V} \left(\frac{\partial V}{\partial T} \right)_P = \frac{R}{P} \frac{T}{V}$$

- (15) Rearranging:

$$\frac{\left(\frac{\partial v}{\partial T}\right)}{\left(\frac{\partial T}{\partial T}\right)} = \frac{RT}{PV}$$

(16) As proven above in steps 8, 9, and 10 and that $1 = \frac{RT}{PV}$:

$$(17) \left(\frac{\partial \ln V}{\partial \ln T}\right)_P = 1$$

(18) If we substitute equation 17 into equation 11:

$$(19) \rho C_p \left(\frac{\partial T}{\partial P}\right)_H = 1 - 1 = 0$$

(20) Since ρ and C_p are non-zero, this means that ∂T must be equal to zero.

(21) **This means that the temperatures at the inlet and exit in a throttling valve are equivalent and therefore the difference in temperature is zero.**

Problem 3B: Review of Ideal Gas and Maxwell Equations

Starting with the enthalpy and internal energy differential equations derived in lecture:

$$(1) dH = C_p dT + (\alpha - \beta) V dP$$

$$(2) dU = C_v dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

Under constant pressure, equation one's second term cancels because $dP = 0$. This leaves only:

$$(3) dH = C_p dT$$

Under constant volume, equation two's second term cancels because $dV = 0$. This leaves only:

$$(4) dU = C_v dT$$

Prove that:

$$(5) C_v \left(\frac{\partial P}{\partial \rho}\right)_S = C_p \left(\frac{\partial P}{\partial \rho}\right)_T$$

Since $\rho = \frac{1}{V}$, taking the derivative leaves: $d\rho = -\frac{1}{V^2} dV$ so we can transform the above equation into:

$$C_v \left.\frac{\partial P}{\partial V}\right|_S = C_p \left.\frac{\partial P}{\partial V}\right|_T$$

This shows that $\left.\frac{\partial P}{\partial V}\right|_T = \left(\frac{\partial P}{\partial \rho}\right)_T$

Since $c_v = \left(\frac{\partial U}{\partial T}\right)_V$ and $C_p = \left(\frac{\partial H}{\partial T}\right)_P$

Using equation 5 and the definitions above, the relationship can be made that:

$$\left(\frac{\partial U}{\partial T}\right)_V \left(\frac{\partial P}{\partial \rho}\right)_S = \left(\frac{\partial H}{\partial T}\right)_P \left.\frac{\partial P}{\partial V}\right|_T$$

After rearranging:

$$\left(\frac{\partial P}{\partial \rho}\right)_S = \frac{\left(\frac{\partial H}{\partial T}\right)_P \left.\frac{\partial P}{\partial V}\right|_T}{\left(\frac{\partial U}{\partial T}\right)_V}$$

After canceling terms:

$$(6) \left(\frac{\partial P}{\partial \rho} \right)_s = - \frac{\left(\frac{\partial H}{\partial V} \right)_P}{\left(\frac{\partial U}{\partial P} \right)_V}$$

Taking equation 3 and dividing each term by dU we find that:

$$(7) (dH = C_p dT) / du = \left(\frac{\partial H}{\partial V} \right)_P = C_p \left(\frac{\partial T}{\partial V} \right)_P$$

Taking equation 4 and dividing each term by dP gives:

$$(8) \left(\frac{\partial U}{\partial P} \right)_V = C_v \left(\frac{\partial T}{\partial P} \right)_V$$

T = T(P, V) Taking the derivative of the temperature function and dividing by dV :

$$(9) \frac{dT}{dV} = \left(\frac{\partial T}{\partial P} \right)_V \cdot \left(\frac{dP}{dV} \right)_T + \left(\frac{\partial T}{\partial V} \right)_P \cdot \left(\frac{dV}{dV} \right)$$

The last term's numerator and denominator cancel.

At a constant temperature, dT = 0.

$$0 = \left(\frac{\partial T}{\partial P} \right)_V \cdot \left(\frac{dP}{dV} \right)_T + \left(\frac{\partial T}{\partial V} \right)_P \text{ the } dP \text{ terms cancel and leave:}$$

$$(10) \left(\frac{\partial P}{\partial V} \right)_T = - \left(\frac{\left(\frac{\partial T}{\partial V} \right)_P}{\left(\frac{\partial T}{\partial P} \right)_V} \right)$$

$$\text{To prove that : } \left(\frac{\partial P}{\partial \rho} \right)_s = - \left(\frac{\left(\frac{\partial H}{\partial V} \right)_P}{\left(\frac{\partial U}{\partial P} \right)_V} \right)$$

$$dU = TdS - PdV$$

U = U(P, V) the derivative leads to:

$$dU = \left(\frac{\partial U}{\partial P} \right)_V \cdot dP + \left(\frac{\partial U}{\partial V} \right)_P dV = TdS - PdV$$

Each term was then divided by dV:

$$(11) \frac{T dS}{dV} = \left(\frac{\partial U}{\partial P} \right)_V \cdot \left(\frac{dP}{dV} \right) + \left[\left(\frac{\partial U}{\partial V} \right)_P + P \right] \cdot \left(\frac{dV}{dV} \right)$$

Assuming constant entropy, dS = 0.

$$(12) \left(\frac{\partial U}{\partial P} \right)_V \cdot \left(\frac{dP}{dV} \right)_S = - \left[\left(\frac{\partial U}{\partial V} \right)_P + P \right]$$

Using H = U + PV

$$(13) dH = Tds + VdP + PdV$$

Taking the derivative of equation 13 at constant pressure, the VdP term = 0 and we have:

$$(14) \left(\frac{\partial H}{\partial V} \right)_P = \left(\frac{\partial U}{\partial V} \right)_P + P$$

Rearrange equation 14:

$$(15) \left(\frac{\partial U}{\partial V} \right)_P = \left(\frac{\partial H}{\partial V} \right)_P - P$$

Rearranging equation 12 shows that:

$$(16) \left(\frac{\partial P}{\partial V} \right)_S = \frac{- \left[\left(\frac{\partial U}{\partial V} \right)_P + P \right]}{\left(\frac{\partial U}{\partial P} \right)_V}$$

The numerator of equation 16 is equal to the negative of equation 14 therefore:

$$(17) \left(\frac{\partial P}{\partial V} \right)_S = \frac{- \left(\frac{\partial H}{\partial V} \right)_P}{\left(\frac{\partial U}{\partial P} \right)_V}$$

Equation 17 proves that: $C_v \left(\frac{\partial P}{\partial \rho} \right)_S = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$

Part 2:

$$(1) C_v \left(\frac{\partial P}{\partial \rho} \right)_S = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

Using the relationship above, calculate the adiabatic speed of sound in the air at ambient temperature assuming that the air behaves as an ideal gas

$$(2) v^2 = \left(\frac{\partial P}{\partial \rho} \right)_S$$

Substitute equation 2 into 1:

$$(3) C_v v^2 = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

Solve for v^2 :

$$(4) v^2 = \frac{C_p}{C_v} \left(\frac{\partial P}{\partial \rho} \right)_T$$

(5) Values at 20 degrees C and 1 bar from engineeringtoolbox.com:

$$T = 293.15 \text{ K}$$

$$C_p = 1.006 \text{ kJ/(kg K)}$$

$$C_v = 0.7172 \text{ kJ/(kg K)}$$

$$R = 287 \text{ J/(kg K)}$$

(6) Assuming ideal gas:

$$PV = RT$$

Substitute $\rho = \frac{1}{v}$ in the relation above.

$$\frac{P}{\rho} = RT$$

Solve for P:

$$(7) P = \rho RT$$

Differentiate equation 7 at a constant temperature with respect to ρ :

$$\left(\frac{\partial P}{\partial \rho} \right)_T = RT$$

Substitute this value into equation 4:

$$v^2 = \frac{c_p}{c_v} (RT)$$

$$v^2 = \left(\frac{1.006 \text{ kJ}/(\text{kg K})}{0.7172 \text{ kJ}/(\text{kg K})} \right) (287 \text{ J}/(\text{kg K})) (293.15 \text{ K})$$

$$v = 343.53 \text{ m/s}$$

According to engineeringtoolbox.com, the speed of sound at 20 °C and 1 bar is 343.1 m/s.

$$\% \text{ error} = \frac{343.53 \frac{\text{m}}{\text{s}} - 343.1 \frac{\text{m}}{\text{s}}}{343.1 \frac{\text{m}}{\text{s}}} = 0.13 \%$$

Problem 3C: Natural Gas Cooling using Refrigerant Propane

Assuming ideal gas behavior.

1. The amount of refrigerant required in KMTA (assume a 90% operating factor)

100 MMSCFD conversion to kmol/s:

100x10 ⁶ ft ³	(0.3048 m) ³	1 day	1 hour
1 day	(1 ft) ³	24 hours	3600 s

$$\dot{V} = 32.77 \text{ m}^3/\text{s}$$

The ideal gas law at STP can be used to calculate the molar flow rate of the methane, $\dot{n}_m$.

$$PV = nRT$$

$$n = (PV)/(RT)$$

$$\dot{n} = \frac{P\dot{V}}{RT} = \frac{\left(\frac{32.7741 \text{ m}^3}{\text{s}} \right) \cdot (1 \text{ atm})}{\left(8.2057 \times 10^{-5} \frac{\text{m}^3 \text{ atm}}{\text{mol K}} \right) (288.7 \text{ K})} = 1383.47 \text{ mol/s} / 1000 = 1.3835 \text{ kmol/s}$$

$$\dot{n}_m = 1.384 \text{ kmol/s}$$

Stream 3 = stream 3 ideal:

Saturated liquid propane

$$T_3 = 35 \text{ °C}$$

$$P_{\text{sat}} = 11.4361 \text{ bar} \quad (\text{using } P_{\text{sat}} \text{ function in calculator})$$

$$\text{Use formula: } h_i^{\text{sat}} = h_{\text{hv}} - \Delta h_{\text{vap}}$$

$$\Delta h_{\text{vap}} = 13860.6708 \text{ kJ/kmol} \quad (\text{from } h_{\text{vap}} \text{ function in calc})$$

$$h_v = h^{\text{IG}} = -103166.9836 \text{ kJ/kmol}$$

$$h_i^{\text{sat}} = -103166.9836 \text{ kJ/kmol} - 13860.6708 \text{ kJ/kmol} = -117027.6544 \text{ kJ/kmol}$$

$$h_i^{\text{sat}} = h_3 = -117027.6544 \text{ kJ/kmol}$$

Valve:

Energy balance:

$$-\Delta H + Q + W = 0$$

No work, no heat transfer to the surroundings.

$$\Delta H = 0 = h_4 - h_3$$

$$h_3 = h_4$$

$$h_4 = -117027.6544 \text{ kJ/kmol}$$

$$T_4 = T_{\text{sat}} = -42.1525 \text{ }^\circ\text{C} \quad (\text{used } T_{\text{sat}} \text{ function in calculator})$$

$$\text{Vapor molar enthalpy} = \text{HIG}(\text{comp1}, -42.1525) = -108364.8961 \text{ kJ/(kmol K)}$$

$$\Delta h_{\text{vap}} = h_{\text{vap}}(\text{comp1}, -42.1525) = 18791.3652 \text{ kJ/(kmol K)}$$

$$\text{Liquid molar enthalpy} = -127156.2613 \text{ kJ/(kmol K)}$$

Stream 4 must be a mixture because the enthalpy is in-between the liquid molar enthalpy and the vapor molar enthalpy.

Stream 4 ideal = stream 4 real

Stream 5:

$$T_5 = 25 \text{ }^\circ\text{C}$$

$$\text{Assume it is all gas. } H_5 = H_{\text{vap}} = \text{HIG}(\text{comp2}, 25) = -74901.9680 \text{ kJ/kmol}$$

$$\dot{n}_m = 1.384 \text{ kmol/s}$$

Stream 6:

$$T_6 = -37 \text{ }^\circ\text{C}$$

$$h_6 = -76998.5652 \text{ kJ/kmol}$$

$$\dot{n}_m = \text{molar flow rate of methane}$$

$$\dot{n}_p = \text{molar flow rate of propane}$$

90 % operating factor means it is operating 90 % of the year.

$$\dot{n}_m * 0.90 = 1.2456 \text{ kmol/s}$$

$$Q_c = \dot{n}_m (h_6 - h_5) = \dot{n}_p (h_4 - h_1)$$

Qc out of the condenser is equal to the Q_H entering the chiller.

$$Q_c = -2611.5215 \text{ kJ/s}$$

$$Q_H = 2611.5215 \text{ kJ/s}$$

$$Q_c = \dot{n}_p (h_4 - h_1) = 2611.5215 \text{ kJ/s}$$

Rearrange to solve for $\dot{n}_p$

$$\dot{n}_p = \left(\frac{\dot{Q}_c}{h_4 - h_1} \right)$$

Stream 1:

Assuming saturated vapor is leaving the chiller and the pressure remains the same, temperature doesn't change in the chiller and only latent heat is removed.

$$T_1 = T_4$$

$$T_1 = -42.1525^\circ\text{C}$$

$$h_{v,1} = h_1 = -108364.8961 \text{ kJ/kmol} \quad \text{using HIG function in calculator}$$

$$h_4 = -117027.6544 \text{ kJ/kmol}$$

$$\dot{n}_p = \left(\frac{\dot{Q}_c}{h_4 - h_1} \right) = 0.3015 \text{ kmol/s propane}$$

KMTA means 1000 metric tons per year.

$$\text{Convert to KMTA} = (0.3015 \text{ kmol/s} * (44.1 \text{ kg/kmol}) * (3600 \text{ s/h}) * (24 \text{ hr/day}) * (365 \text{ day/yr}) * (1 \text{ tons/1000 kg})) / 1000 = 419.307 \text{ KMTA}$$

The amount of refrigerant required is approximately 420 KMTA of propane.

2. The duties (in MW) of the chiller and the condenser.

Chiller:

$$\text{Energy balance: } -\Delta H + Q + W = 0 \quad W=0 \quad Q = \Delta H$$

$$Q_c = (h_1 - h_4) \dot{n}_p$$

$$Q_c = 2209.8404 \text{ kJ/s} = 2209.8404 \text{ kW} = \mathbf{-2.21 \text{ MW}}$$
 is the duty for the chiller using ideal enthalpies

Duty required by the chiller is 2.21 MW.

Condenser:

$$\text{Energy balance: } -\Delta H + Q + W = 0 \quad W=0 \quad Q = \Delta H$$

$$Q_c = (h_3 - h_2) \dot{n}_p \quad \text{so I need to solve for } h_2$$

To find stream 2 we need to solve for stream 1.

The saturated vapor that exits the chiller stays at the same pressure. So $P_4 = P_1$. Then we do a balance around the compressor since compressors are ideally isentropic. $S_1 = S_2$.

$$S_1 = -286.4698 \text{ kJ/(kmol K)}$$

Use sig function in excel to find the vapor molar entropy in stream 2 using a guess value for the temperature. Then I used secant method by setting $S_2 = S_1$ by changing the temperature of stream 2.

$$T_2 = 38.9218^\circ\text{C}$$

Then using the hlg function, the enthalpy in stream 2 is:

$$h_2 = -102867.1489 \text{ kJ/kmol}$$

$$Q_c = (h_3 - h_2) \dot{m}_p = (-117027.6544 \text{ kJ/kmol} - 102867.1489 \text{ kJ/kmol}) * 0.3015 \text{ kmol/s}$$

$$Q_c = -4268.9018 \text{ kJ/s} = \text{kW} = -4.27 \text{ MW}$$
 this is the duty required by the condenser using ideal enthalpies

Duty required by the condenser 4.27 MW.

3. The power (MW) requirement in the compressor assuming:

a. 75% isentropic efficiency

Energy balance for compressor:

$$-\Delta H + Q + W = 0 \quad \text{Assuming } Q = 0 \text{ the energy balance of a compressor becomes:}$$

$$W = \dot{n}_p (h_2 - h_1) \quad \text{using ideal enthalpies since we based } h_2 \text{ on the compressor being isentropic.}$$

The compressor is 75 % isentropic so it is necessary to divide this result by 0.75. Division is necessary because the work required by the compressor will be higher than the ideal work.

$$W_i = (0.3015 \text{ kmol/s} (-102867.1489 \text{ kJ/kmol} - 108364.8961 \text{ kJ/kmol})) / 0.75 = 2209.8404 \text{ kW}$$

$W_{\text{compressor}} = 2.21 \text{ MW}$ is the power requirement of the compressor assuming it has 75% isentropic efficiency

b. 75 % polytropic efficiency

From our lecture notes in chapter 2 the polytropic work is defined as:

$$W_s = \frac{\gamma}{\gamma - 1} R T_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right]$$

Using engineeringtoolbox.com, $C_p = 0.0634$ and $C_v = 0.05363$ for propane means that $\gamma = \frac{C_p}{C_v} = 1.18$

$$T = -42.1525 \text{ }^\circ\text{C} + 273.15 = 230.998 \text{ K}$$

$$W_s = \left(\frac{1.18}{1.18 - 1} \right) \left(8.3145 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \right) 230.998 \text{ K} \left[\left(\frac{11.4361 \text{ bar}}{1 \text{ bar}} \right)^{\frac{1.18 - 1}{1.18}} - 1 \right] * \dot{n}_p / 0.75$$

$W_s = 2278.76 \text{ kJ/s} = 2.28 \text{ MW}$ is the power requirement of the compressor assuming 75% polytropic efficiency

4. Cop for isentropic system:

$$\text{COP}_{\text{isentropic}} = \frac{|Q_c|}{|Q_H - Q_c|}$$

Q_H is the heat leaving the natural gas and entering the condenser.

Q_c is the heat leaving the condenser

Q_H stays the same.

Finding the Q_c for isentropic efficiency of 75%:

$$h_2^{\text{real}} = (h_2^{\text{ideal}} - h_1^{\text{ideal}}) / 0.75 + h_1^{\text{ideal}}$$

$$h_2^{\text{ideal}} = -102867.1489 \text{ kJ/kmol}$$

$$h_1^{\text{ideal}} = -108364.8961 \text{ kJ/kmol}$$

By substituting these values into the above formula:

$$h_2^{\text{real}} = -101034.5665 \text{ kJ/kmol}$$

$$Q_c = (h_3^{\text{ideal}} - h_2^{\text{real}}) \dot{n}_p = -4821.3618 \text{ kJ/s}$$

$$\text{COP}_{\text{isentropic}} = |(4821.3618 \text{ kJ/s}) / (2611.5215 \text{ kJ/s} - 4821.3618 \text{ kJ/s})| = 2.18$$

$\text{COP}_{\text{isentropic}} = 2.18$

Finding the Q_H for polytropic efficiency of 75%:

$$W^{\text{real}} = \dot{m}(h_{2r} - h_1)$$

$$h_{2r} = (W^{\text{real}}) / \dot{m} + h_1$$

$$h_{2r} = -100805.9511 \text{ kJ/kmol}$$

$$Q_c = (h_3^{\text{ideal}} - h_2^{\text{real}}) \dot{m}_p = -4890.2815 \text{ kJ/s}$$

$\text{COP}_{\text{polytropic}} = \left \frac{Q_c}{Q_H - Q_c} \right = 2.1460$
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$$5. \text{ Max COP} = \frac{|T_C|}{|T_H - T_C|}$$

$$T_c = T_1 = -42.1525 \text{ }^\circ\text{C} + 273.15 = 230.998 \text{ K}$$

$$T_H = T_2 = 38.9218 \text{ }^\circ\text{C} + 273.15 = 312.072 \text{ K}$$

Max COP = 2.849
